Beyond GDP: Country Economic & Public Health Analysis

An interactive Tableau project exploring the relationships between economic prosperity, demographics, and public health indicators across countries and entities.

Project Overview

This project analyzes cross-country economic and public health indicators to explore how economic prosperity relates to population well-being.

The main objective is to move beyond total GDP and investigate whether higher income per capita is associated with better public health outcomes, particularly life expectancy and infant mortality.

The project uses interactive Tableau dashboards and data storytelling to communicate analytical findings and identify patterns, correlations, and outliers.

Business Questions

- How does GDP per capita provide a different perspective from total GDP?
- Is higher economic prosperity associated with higher life expectancy?
- Is GDP per capita associated with lower infant mortality?
- How are economic growth rates distributed across countries?
- Do life expectancy distributions differ across income groups?
- Which countries or entities represent notable economic or health-related outliers?

Dataset

The cleaned dataset contains:

- 263 country and entity records.
- 45 fields, including country names and 44 economic, demographic, and health indicators.
- 262 records after excluding the European Union aggregate from individual-entity comparisons.

Data Source: [Insert Original Dataset Source URL]

Important: The dataset does not contain a Year or Date field. Therefore, this project focuses on cross-sectional comparisons and statistical relationships rather than time-series analysis.

Tools & Technologies

- Tableau Public / Tableau Desktop
- Calculated Fields
- Parameters
- Dashboard Actions
- Interactive Filters
- Data Visualization
- Data Cleaning and Validation
- Tableau Storytelling

Project Structure

Dashboard 1 — Global Overview

Provides an interactive overview of economic and development indicators.

Key components:

- Dynamic geographical map.
- Dynamic country ranking bar chart.
- KPI cards for median GDP per capita, median life expectancy, median infant mortality, and complete core metric records.

- Dynamic Metric Selector.

Dashboard 2 — Prosperity & Public Health

Explores the relationship between economic prosperity and public health.

Key components:

- GDP per capita vs. life expectancy scatter plot.
- GDP per capita vs. infant mortality scatter plot.
- Country economic and health comparison table.
- Selected-country highlighting through Set Actions.

Dashboard 3 — Demographics & Economic Patterns

Analyzes economic growth, income-group distributions, and total economic size.

Key components:

- GDP growth rate histogram.
- Life expectancy box-and-whisker plot by GDP per capita groups.
- GDP treemap by country/entity.

Visualizations

The project includes six primary visualization types:

1. Filled Geographical Map.
2. Horizontal Bar Chart.
3. Scatter Plot.
4. Histogram.
5. Box-and-Whisker Plot.
6. Treemap.

Additional KPI cards and comparison tables provide further analytical detail.

Interactivity

The Tableau workbook incorporates:

- Dynamic Metric Selector parameter.
- Multiple calculated fields.
- Interactive country filters.
- Dashboard Filter Actions.
- Dashboard Highlight Actions.
- Set Actions for selected-country comparisons.
- Interactive Tableau Story navigation.

Key Analytical Findings

The analysis reveals several descriptive relationships:

- GDP per capita and life expectancy show a positive association (Pearson correlation approximately +0.61).
- GDP per capita and infant mortality show a negative association (approximately -0.59).
- Life expectancy and infant mortality show a strong negative association (approximately -0.89).

These calculations use approximately 225 pairwise-complete observations for each comparison.

These findings describe statistical associations and should not be interpreted as evidence of causation.

Data Preparation & Methodology

- Removed non-data metadata rows.
- Validated numerical field types.

- Preserved missing values as NULL instead of replacing them with zero.
- Excluded the European Union aggregate from individual-entity comparisons.
- Created project-defined GDP per capita groups.
- Used median-based KPI calculations where appropriate.
- Used statistical trend lines to explore relationships between economic and health indicators.

Limitations

- The dataset does not include a reference year or date.
- Time-series trends cannot be established.
- The dataset includes territories and dependencies in addition to countries.
- Missing values vary significantly across indicators.
- Correlation does not establish causation.
- GDP per capita alone cannot explain differences in public health outcomes.

Repository Contents

- `workbook/` : Packaged Tableau workbook (.twbx).
- `images/` : Dashboard and Story screenshots.
- `data/` : Cleaned dataset, where redistribution is permitted.
- `README.md` : Project documentation.

How to Explore the Project

Interactive Version: [Insert Tableau Public URL]

Workbook: Download `Country_Comparison_Storytelling.twbx` from the workbook folder and open it using Tableau Desktop or Tableau Public.

Dataset: See the data folder or the original dataset source linked above.

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